

Nahee **Kim**

M.S. Student in Data Science · Model Interpretability · Model Reliability · Representation Dynamics

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Research Profile

My research focuses on understanding neural networks through their internal dynamics and representations. In RiemannMamba, I introduced geometry-aware internal time to characterize hidden-state evolution and studied its relationship to sample difficulty, error propensity, and reliability under distribution shift. I am currently investigating whether learned state trajectories in selective state-space models exhibit systematic variational structure under an input-dependent internal clock, and how deviations from this structure relate to model failure. More broadly, I am interested in mechanistic interpretability, model evaluation, and identifying the internal mechanisms that drive robust or brittle model behavior.

Education

Inha University — *Incheon, Republic of Korea*

M.S. in Data Science — *Sep. 2025 – Feb. 2027 (Expected)*

- Expected to complete the M.S. degree in three semesters through early graduation
- Research: Mechanistic interpretability, representation dynamics, model reliability, selective state-space models
- Advisor: Prof. Ji Hyun Nam
- GPA: 4.13 / 4.5

Inha University — *Incheon, Republic of Korea*

B.S. in Electronic Engineering — *Mar. 2019 – Aug. 2025*

- GPA: 3.80 / 4.5

Selected Research & Publications

RiemannMamba: Geometry-Aware Internal Time in Selective State Space Models — *Under review at NeurIPS 2026*

Sole First Author — *Aug. 2025 – Apr. 2026*

- **Internal mechanism:** Reinterpreted Mamba's input-dependent discretization step Δ as a state-dependent geometric factor and used it to define a geometry-aware internal time τ, providing an analysis framework for characterizing how selective dynamics modulate hidden-state evolution across different inputs.
- **Reliability and evaluation:** Quantified how internal time varies with sample difficulty and prediction errors, and evaluated τ as a complementary white-box reliability signal for selective prediction across clean, distribution-shifted, and corrupted inputs.
- **Interpretability:** Developed gradient-free τ-CAM to attribute accumulated internal-time changes back to spatial input regions, enabling visualization of where and how strongly an input induces hidden-state evolution without relying on gradient-based saliency.

TST-Mamba: Plug-and-Play Selective State-Space Denoising for Non-Line-of-Sight Imaging — *Under review at Neurocomputing*

Sole First Author — *Jan. 2025 – Jul. 2026*

- **Method:** Developed a Mamba-based spatio-temporal denoising architecture and a real-measurement-guided noise simulation framework for transient imaging.
- **Architecture:** Designed and formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder for NLOS transients, combining an undercomplete temporal bottleneck, single-pass spatial processing in latent space, and skip–latent fusion to preserve reconstruction-critical temporal details.
- **Generalization:** Evaluated robustness across simulated, unseen-category, and real measurements, demonstrating plug-and-play transfer across reconstruction pipelines.

Driver Handover Readiness Estimation via Bio-Vehicle Signal Fusion and Safety-Oriented Re-decision — *IEIE Summer Annual Conference 2026*

Lead Researcher · Accepted Conference Paper — *Jun. 2026*

- **Project leadership and mentorship:** Conceived the project and led it end-to-end, while supervising an undergraduate researcher through problem formulation, framework design, experimental evaluation, result analysis, and manuscript preparation.
- **Methodology:** Developed a bio-vehicle signal fusion framework combining ECG-derived driver stress estimates with vehicle-state information, and designed a reward-based re-decision policy to reduce unsafe handover approvals.
- **Validation and presentation:** Evaluated the framework through driver-held-out experiments on WESAD and OpenDriver data; the paper was selected for the conference's best-paper session and presented as a poster in Jeju, Republic of Korea.

Air Purifier Utilizing Tesla Compressor and Cyclone Dust Collector — *Technology and Education · KSME*

KCI Journal Paper — *2024*

- **Research contribution:** Implemented motor control for the Tesla compressor and independently conducted the experimental analysis and manuscript writing.

Additional Research Experience

SocialWise: Augmented-Reality-Based Social Communication Support for Children with Autism Spectrum Disorder — *Inha University*

Undergraduate Research Assistant — *2024*

- **Human-centered assistive AI:** Contributed to the development of an augmented-reality-based system designed to interpret social interaction cues and support social communication for children with autism spectrum disorder.
- **Speech data collection:** Conducted speech-utterance data collection with participants with autism spectrum disorder for the development of the assistive communication system.

Selected Projects

Cross-Lingual Voice-Conversion Augmentation for Low-Resource Neural Vocoding — *Inha University*

Capstone Design Team Project — *Dec. 2024*

- **Low-resource data augmentation:** Designed three data-scarcity settings retaining 50%, 25%, and 12.5% of the original LJSpeech data, and augmented the missing portions through FreeVC-based voice conversion using speaker references from one, three, and seven language backgrounds.
- **Model training and evaluation:** Trained a full-data baseline and three augmented-data MelGAN vocoders for 3,000 epochs, and compared their convergence and speech reconstruction quality using L1 loss, waveforms, and mel-spectrograms; the 50% and 25% settings approached the full-data baseline while converging faster during early training.

Teaching Experience

Object-Oriented Programming — *Inha University*

Teaching Assistant — *Spring 2026*

- Assisted students with object-oriented programming concepts, programming exercises, and the implementation and debugging of course assignments.

Image Data Analysis — *Inha University*

Teaching Assistant — *Spring 2026*

- Supported coursework on image-data processing and analysis, helping students understand computational methods, experimental workflows, and the interpretation of image-based results.

Physics Laboratory — *Inha University*

Teaching Assistant — *Fall 2025*

- Led full undergraduate physics laboratory sessions, including explaining experimental principles and procedures, supervising measurements and equipment use, guiding data analysis, and evaluating students' experimental work.

High School Physics — *Mirae Injae Academy*

Physics Instructor — *May 2023 – Apr. 2024*

- Taught high school students fundamental physics concepts and problem-solving methods, explaining mathematical and physical principles in an accessible and structured manner.

Honors & Awards

Oct. 2023	**Bronze Award,** 13th National Student Design Competition (Tesla-Compressor Cyclone Collector)	*KSME*
Oct. 2023	**Excellence Award,** 15th Creative Design Competition (Multi-Phase Hybrid Generator)	*I-DREAM*
Sep. 2020	**Bronze Award,** 2020 Capstone Design Competition (Automatic Nozzle Changer for 3D Printers)	*Inha University*
Jul. 2020	**Grand Prize,** 2020 Engineering Design Club Competition (Automatic Nozzle Changer for 3D Printers)	*Inha Univ. ICEE*

Skills & Research Expertise

Programming	Python, C, C++
Deep Learning	PyTorch; custom architecture design and training; Mamba/SSMs; CNNs; spatio-temporal models
Model Analysis	Hidden-state tracing; internal-dynamics analysis; representation geometry; selective prediction
Model Evaluation	ImageNet-scale evaluation; corruption and distribution-shift robustness; reliability analysis
Research Computing	Linux, Bash, Git/GitHub; GPU experimentation; mixed-precision training; reproducible pipelines
Research Interests	Mechanistic interpretability, representation dynamics, and model reliability
Languages	Korean (native), English (working proficiency)

References

Prof. Ji Hyun Nam	Assistant Professor and Department Chair, Department of Statistics and Data Science, Inha University | namjihyun@inha.ac.kr | M.S. Advisor
Prof. Kyounghee Baek	Professor, School of Law, Inha University | khbaek@inha.ac.kr | Academic Referee (AI Ethics & Law Coursework)